

Muhammad Mohad Umer Chaudhary

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EDUCATION

Ghulam Ishaq Khan Institute (GIKI) | Bachelor of Science – Mechanical Engineering

2022 – 2026

- Relevant coursework: System Controls & Dynamics, Fluid Mechanics, Mechanics of Solids, AI for Mechanical Engineers, Electric Vehicles
- Position Holder for **Best Final Year Project** 2026 at **MTME'26 conference**

The City School, Islamabad | A-Level & O-Level

2018 – 2022

- Elected Campus Head Boy; served on student council for 8 consecutive years
- Commendable academic results in O-Level and A-Level examinations
- Led Al-Khidmat Foundation fundraising program as Head Boy

PROFESSIONAL EXPERIENCE

Government of Punjab – Agriculture Engineering Department | Mechanical Engineering Intern

Jun – Jul 2025

- Applied mechanical engineering principles to agricultural machinery inspection and irrigation system analysis
- Gained practical experience in equipment optimization for sustainable and efficient farming practices
- Assisted in maintenance scheduling and performance evaluation of farm machinery

WASA – Water and Sanitation Agency | Engineering Intern

Jul – Aug 2025

- Developed understanding of municipal water supply infrastructure and sanitation systems
- Observed and documented maintenance procedures for pumps, motors, and water distribution equipment
- Supported operational teams in identifying performance inefficiencies in water supply networks

Aga Khan Foundation | Human Resource Intern

Jul – Aug 2024

- Supported talent acquisition processes including screening, scheduling, and onboarding coordination
- Contributed to employee engagement initiatives and streamlined HR operational workflows
- Assisted in documentation, reporting, and coordination of organizational HR programs

Al-Khidmat Foundation | Head of Fundraising

Jan 2020

- Organized and led community fundraising campaigns as Campus Head Boy to support financially distressed individuals
- Coordinated volunteers, managed logistics, and ensured successful execution of the initiative

FINAL YEAR PROJECT

Flexible Thin-Film Wearable Heater (1st Position at MTME'26 and GIKI Openhouse'26)

2025 – 2026

- Developed flexible heating elements using screen-printed silver and carbon conductive pastes for wearable textile applications
- Evaluated electro-thermal performance, power efficiency, and mechanical durability under repeated flexing conditions
- Keywords: flexible electronics, conductive paste, screen printing, wearable technology, electro-thermal analysis

ACADEMIC PROJECTS

- Design and Finite Element Analysis (FEA) of Spur Gear Assembly using SolidWorks
- Computational Flow Analysis across Cross-Section of Shell-and-Tube Heat Exchanger using ANSYS
- Nature-Inspired Airfoil Design using Generative AI and Computational Fluid Dynamics (CFD)
- Business Impact Analysis: Assessing Operational Suspension Due to Economic Challenges

TECHNICAL SKILLS

CAD / CAE Software: SolidWorks, AutoCAD, ANSYS Workbench, COMSOL Multiphysics, Simulink, MATLAB

Engineering Competencies: Finite Element Analysis (FEA), CFD, Thermal Analysis, Mechanical Design, System Dynamics

Productivity Tools: Microsoft Word, Excel, PowerPoint

Soft Skills: Leadership, Team Coordination, Fundraising Management, Communication, Problem-Solving